

22/07/24

① How well did Ram perform in his English
classwork compared to the other 50 students?

Sol

Given,

Ram scored $= 70$

Mean $= 60$

S.D $= 15$

W.K.T $z = \frac{x - \mu}{\sigma} = \frac{70 - 60}{15} = \frac{10}{15} = 0.67$

Answer $= 0.67$

② Comparing Quiz Performances.

Sol

Quiz 1:

Given: $x_1 = 80$, $\mu_1 = 70$, $\sigma_1 = 15$

$z_1 = \frac{80 - 70}{15} = \frac{10}{15} = 0.67$

Quiz 2:

Given: $x_2 = 75$, $\mu_2 = 54$, $\sigma = 12$

$z_2 = \frac{75 - 54}{12} = \frac{21}{12} = 1.75$

③ Interquartile Range (IQR) for First 10 Prime Numbers;

Sol:

1) Input: Str1 = "greek", str2 = "greek"

2) Input: Str1 = "cat", str2 = "cat"

3) Input: Str1 = "Sunday", str2 = "Saturday"

First 10 Prime numbers:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29

→ lower half: 2, 3, 5, 7, 11, ~~13~~ → $O_1 = 5$

→ Upper half: 13, 17, 19, 23, 29 → $O_3 = 19$

$$IQR = O_3 - O_1 = 19 - 5$$

$$= 14$$